Research on Quad-Frequency PPP-B2b Time Transfer

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Carrier phase time transfer which is a crucial technique in universal time coordinated (UTC) calculation is implemented through precise point positioning (PPP). Since August 2020, the Beidou global navigation satellite system (BDS-3) has provided users with the precise satellite product which is an essential external input in the PPP implementation, named the PPP-B2b product, through three geostationary earth orbit (GEO) satellites instead of a network in the Asia-Pacific area. The PPP-B2b product can be considered to solve the instability problem caused by network interruption in traditional PPP time transfer. Currently, the fact that the PPP-B2b time transfer using dual-frequency (DF) ionosphere-free combination can achieve sub-nanosecond accuracy has been proven. Considering the BDS-3 can provide users with a wide range of frequency signals for PPP; meanwhile, the multi-frequency PPP will improve the accuracy of time transfer and accelerate the convergence. This improvement can be attributed to an increase in the number of observation equations due to the utilization of multiple frequencies. To promote the application of real-time PPP-B2b time comparison in UTC calculation, a quad-frequency (QF) PPP-B2b time transfer model is proposed and investigated. Compared to DF PPP-B2b time transfer, the accuracy of the QF time transfer model was verified from long-baseline time links and zero-baseline common clock difference (CCD). Results showed that the QF PPP-B2b time transfer had smoother CCD results and fluctuated within 0.1 ns, compared to the DF PPP-B2b model. Taking the PPP time comparison using the GBM product as a reference, the results for all long-baseline links show that the residuals of the QF PPP-B2b time comparison truly fluctuate within 1 ns.

Background

Global navigation satellite system (GNSS) can provide real-time positioning, navigation and time services for users worldwide. Currently, GNSS time transfer mainly includes common-view (CV), all-in-view (AV), and GNSS carrier phase time transfer. CV time transfer requires both users to observe the same satellites, imposing limitations on the distance of the time transfer link. AV, on the other hand, addresses the distance limitation of CV. However, AV relies on low-precision pseudorange observations, resulting in reduced accuracy in time transfer. GNSS carrier phase time transfer through PPP [1] has the highest accuracy among the three methods without distance limitation. Real-time PPP requires external provision of real-time precise satellite orbit and satellite clock offset. The GNSS analysis center provides precise satellite files for researchers to study and verify PPP algorithms. The standard deviation (STD) value of post-process carrier phase time transfer using the GBM product which is provided by The GeoForschungZentrum (GFZ) analysis center with a two-day delay can reach 0.3 ns [2]. Moreover, since 2013, the international GNSS service (IGS) has provided real-time precise satellite products via a network, contributing to the progress and development of PPP. PPP time transfer using the real-time product can achieve 0.5 ns accuracy [3]. However, the acquisition of precise product over a network not only introduces limitations to real-time PPP, but also makes the time transfer unstable because of network interruptions. To mitigate the impact of network interruptions on PPP time transfer, the BDS-3 has provided precise satellite products called the PPP-B2b product through three GEO satellites [4]. Compared with another service broadcast by GEO satellites, satellite-based augmentation system (SBAS), the PPP-B2b product have higher accuracy and shorter update intervals [5]. Currently, the fact that DF PPP-B2b time transfer can reach sub-nanosecond accuracy has been proven [6],[7]. A highly reliable, network-independent time transfer will be feasible, which has led to an increasing number of researchers to study the PPP-B2b time transfer in terms of precision and stability. Given that the BDS-3 can provide users with a wide range of frequency signals for PPP and the increased observation information will accelerate PPP convergence and improve the accuracy of time transfer, QF PPP-B2b time transfer was studied.

In this study, a new QF PPP-B2b time transfer model was investigated. To take full advantage of the BDS-3 frequency signals to comprehensively assess the accuracy of QF PPP-B2b time transfer model, the BDS-3 newly broadcasted

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signals (B1C and B2a) were used alongside the existing B1I and B3I signals. To ensure the reliability of the results, taking the GBM product as a reference, the accuracy of QF PPP-B2b time transfer was verified from long-baseline time links and zero-baseline CCD. The QF PPP-B2b time transfer calculation involved several stations, including two international atomic time (TAI) keeping laboratories, namely the Telecommunication Laboratories (TL) and the National Time Service Center (NTSC), as well as an external atomic clock IGS station (USUD).

Method

Real-time precise satellite position and satellite clock offset can be obtained using ephemeris and the PPP-B2b product [4]. Moreover, the PPP-B2b product also provides PPP user with differential code bias product to correct uncalibrated code delay (UCD) in pseudorange observation equation.

PPP Model

PPP is an absolute positioning technology that achieves high-precision positioning through models, empirical formulas, and parameter estimation methods to process errors, employing a single receiver to receive GNSS satellite signals to obtain the raw carrier phase and pseudorange observations. The basic PPP observation equations can be written as (1) and (2):

$$P_{r,j}^{s} = \rho + c\left(dt_r - dt^s\right) + T_r^s + I_{r,j}^s + c\left(d_{r,j}^s - d_j^s\right) + \varepsilon_{r,j}^s \quad (1)$$

$$L_{r,j}^{s} = \rho + c\left(dt_r - dt^s\right) + T_r^s - I_{r,j}^s + \lambda_j^s\left(N_{r,j}^s + b_{r,j}^s - b_j^s\right) + \xi_{r,j}^s \quad (2)$$

in which L and P denote raw carrier phase and pseudorange observations obtained directly through the receiver, respectively. j is frequency numbering of observation. ρ refers to the geometric distance from receiver r to satellite s. c is a constant of 299792458 m/s. I denote ionospheric delay and T refers to tropospheric delay. dt_r and dt^s denote receiver clock offset and satellite clock offset, respectively. d and b refer to UCD and uncalibrated phase delay (UPD), respectively. N refers to the integer ambiguity. ε and ξ denote the noise in observation measurement. Table 1 lists the main errors and error processing strategies of PPP.

Table 1 – Errors and Processing Strategies of PPP

Errors	Strategies
Satellite orbit	PPP-B2b orbit products
Satellite clock	PPP-B2b clock products
UCD	PPP-B2b products
Tropospheric delay	Model [8],[9] and parameter estimation
Ionospheric delay	Model [10] or parameter estimation
Phase wind-up	Model [11]
Relativistic effect	Model [12]
Satellite and receiver antenna phase center	"atx" file provided by IGS
Receiver coordinates	Parameter estimation
Receiver clock	Parameter estimation
Phase ambiguity	Parameter estimation

Quad-Frequency PPP-B2b Model

For convenience, (3) is defined, and f denotes frequency value involved in the calculation. α and β are DF ionospheric-free (IF) combination coefficient:

$$\alpha_{ij} = \frac{f_i^2}{f_i^2 - f_j^2}, \beta_{ij} = -\frac{f_j^2}{f_i^2 - f_j^2} \quad (3)$$

Considering that the navigation signals provided by the BDS-3 include B1I, B3I, B1C, and B2a which are numbered as 2, 6, 1, and 5, a quad-frequency PPP-B2b time transfer model is developed. Meanwhile, a B3I hardware delay is introduced by the BDS-3 PPP-B2b clock product, which can be written as (4) and corrected using the PPP-B2b differential code bias product. After correcting the B3I hardware delay, the QF PPP-B2b model can be expressed as (5) and (6):

$$c \cdot dt_{B2b}^s = c \cdot dt^s + c \cdot d_6^s \quad (4)$$

$$\begin{cases} P_{r,IF_{26}}^s = \rho + c \cdot dt_{r,IF_{26}} + T_r^s - c \cdot dt_{B2b}^s + \varepsilon_{r,IF_{26}}^s \\ P_{r,IF_{15}}^s = \rho + c \cdot dt_{r,IF_{26}} + \theta_{isb} + T_r^s - c \cdot dt_{B2b}^s + \varepsilon_{r,IF_{15}}^s \\ L_{r,IF_{26}}^s = \rho + c \cdot dt_{r,IF_{26}} + T_r^s - c \cdot dt_{B2b}^s + \bar{\lambda}\bar{N}_{r,IF_{26}}^s + \xi_{r,IF_{26}}^s \\ L_{r,IF_{15}}^s = \rho + c \cdot dt_{r,IF_{26}} + T_r^s - c \cdot dt_{B2b}^s + \bar{\lambda}\bar{N}_{r,IF_{15}}^s + \xi_{r,IF_{26}}^s \end{cases} \quad (5)$$

$$\begin{cases} c \cdot dt_{r,IF_{26}} = c \cdot dt_r + \alpha_{26} d_{r,2}^s + \beta_{26} d_{r,6}^s \\ \theta_{isb} = \alpha_{15} d_{r,1}^s + \beta_{15} d_{r,5}^s - \alpha_{26} d_{r,2}^s - \beta_{26} d_{r,6}^s \end{cases} \quad (6)$$

$$\rho = \rho_0 + \mu_r^s \cdot \Delta x \quad (7)$$

In (5), $P_{r,IF_{ij}}^s$ and $L_{r,IF_{ij}}^s$ refer to the combined IF pseudorange and carrier phase observations at frequencies i and j after correcting UCD using the PPP-B2b product, respectively. $\bar{\lambda}\bar{N}$ denotes the floating ambiguity. ρ can be linearized. ρ_0 refers to the geometric distance using the last estimated receiver coordinates. μ_r^s denotes the unit vector between receiver r to satellite s. Δx refers to the receiver coordinates vector increments. Therefore, after correcting satellite clock offset dt_{B2b}^s through the PPP-B2b product, the QF PPP-B2b time transfer estimated parameter vector $\bar{X}$ is:

$$\bar{X} = \begin{bmatrix} \Delta x & dt_{r,IF_{26}} & T_r^s & \theta_{isb} & \bar{\lambda}\bar{N}_{r,IF_{26}}^s & \bar{\lambda}\bar{N}_{r,IF_{15}}^s \end{bmatrix}. \quad (8)$$

PPP-B2b Time Transfer

The principle of time transfer using the PPP-B2b product is shown as Fig. 1. Fig. 2 illustrates a simplified PPP processing workflow. Meanwhile, Table 2 lists QF PPP-B2b time transfer processing strategies. As far as the acquisition of precise products is concerned, GEO satellites are used in PPP-B2b time transfer instead of a network to broadcast real-time data, compared to the traditional PPP time transfer. The receiver clock offset obtained by PPP is the deviation of the local time t_{loc} compared to the GNSS reference time t_{ref}. The time links involved in time transfer eliminate the GNSS reference time t_{ref} through receiver clock offset difference to achieve high precision time transfer, which is written as:

$$\Delta t = dt_{r,A} - dt_{r,B} = \left(t_{loc,A} - t_{ref}\right) - \left(t_{loc,B} - t_{ref}\right) = t_{loc,A} - t_{loc,B} \quad (9)$$

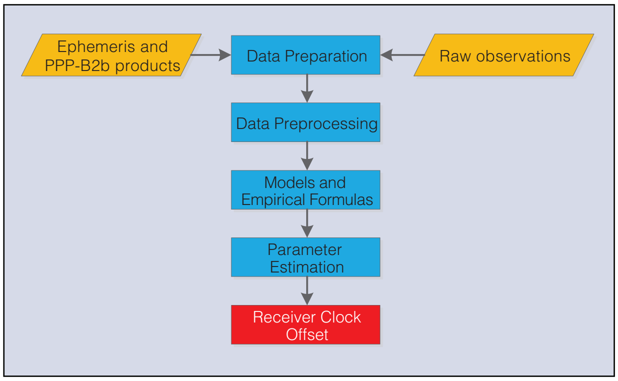

Fig. 2. Simplified PPP processing flow.

Results and Discussion

The zero-baseline CCD [13],[14] experiment can reflect the noise level of time transfer. The receivers involved in the zero-baseline time transfer are linked to a shared GNSS antenna, and the baseline length is set to zero. QF PPP-B2b model was validated by zero-baseline CCD and long-baseline time links. Fig. 3 shows geographic location of two long-baseline time links, and Table 3 lists information of receivers involved in zero-baseline CCD and long-baseline time links. NTSC and TL are TAI timekeeping laboratories, while IGS station USUD simply uses an external atomic clock as the clock source for the GNSS receiver. GNSS receivers, TLM2, NTTS and NT07, are connected to the UTC(k). Taking time comparison using the GBM product as a reference, QF and DF PPP-B2b time comparison were implemented to verify the improvement of PPP-B2b time transfer accuracy by increasing frequency. The observations participating zero-baseline CCD were used for 12 d from day of year (DoY) 85 to 96 in 2022, while those from DoY 163 to 171 in 2022 were used for long-baseline time comparison. Observations can be obtained through

Table 2 – QF PPP-B2b time comparison strategies

Item	Strategies
Navigation system	BDS-3
Raw observations	B1C, B2a, B1I, B3I
Satellite orbit and clock	Ephemeris and the PPP-B2b product
Estimator	Kalman filter
Processing interval	300 s
Pseudorange UCD	Corrected: the PPP-B2b product
Tropospheric delay	Dry delay: Saastamoinen model [8]; Wet delay: estimated as a random walk noise; GMF model [9] is used as the mapping function.
Receiver clock	Estimated: white noise
Receiver coordinates	Estimated: constants (static)
Phase ambiguity	Estimated: floating point

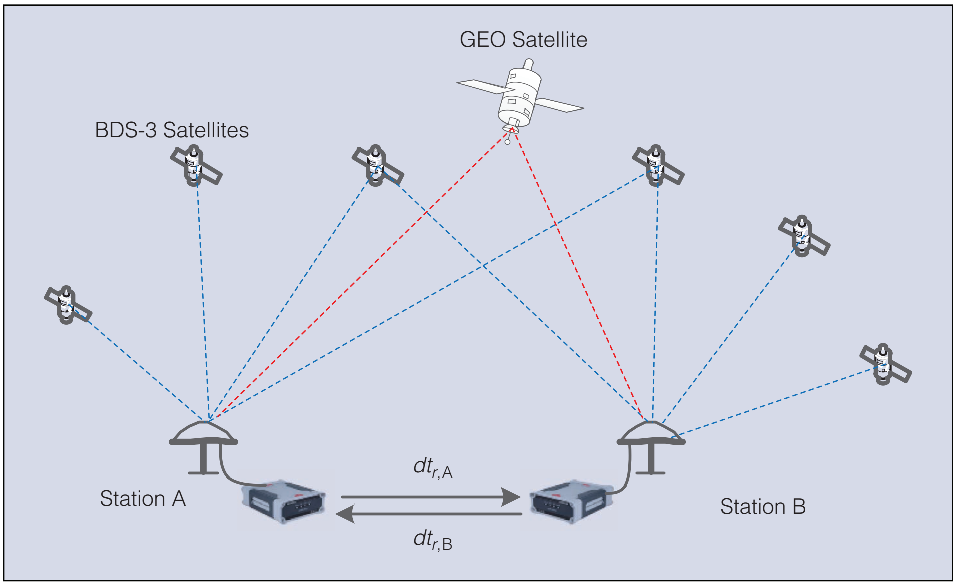

Fig. 1. Principle of PPP-B2b time transfer.

https://cddis.nasa.gov/, and the PPP-B2b product can be buffered by a NovAtel OEM729 receiver equipped with a PPP-B2b module.

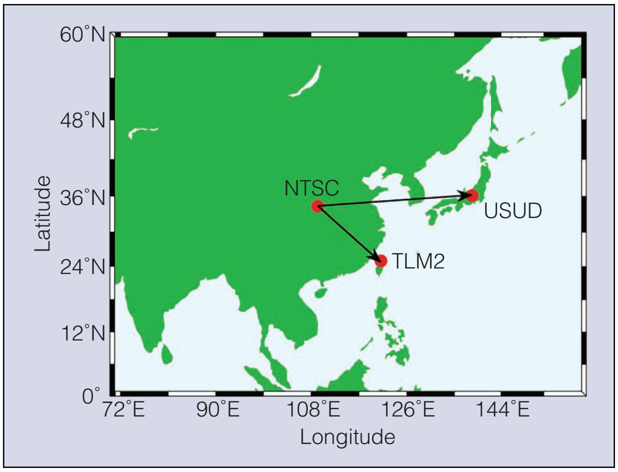

Fig. 3. The geographic distribution of time comparison links stations.

Table 3 – Detailed information of time links receivers

Station/ ID	Organization	GNSS Receiver	Clock
NTTS	NTSC	SEPT POLARX5	EXTERNAL
NT07	NTSC	SEPT POLARX5	EXTERNAL
TLM2	TL	SEPT POLARX5	EXTERNAL
USUD	IGS	SEPT POLARX5	EXTERNAL H-MASER

PPP-B2b CCD

The result of zero-baseline CCD between NTTS and NT07 with a common antenna and external clock using QF and DF PPP-B2b model was displayed in Fig. 4. The STD value of the CCD result was calculated. The satellite products are the same in PPP solution, and only the processing model is different. Ideally the CCD result should be a horizontal line, but the presence of noise makes the result sequence fluctuate. The smoothness of the curve reflects the level of PPP-B2b model noise. Results show that QF PPP-B2b time transfer has smoother CCD results and fluctuates within 0.1 ns, compared to the DF PPP-B2b model.

Long-baseline PPP-B2b Time Comparison

Fig. 5 displays the results of the QF NTSC-TLM2 PPP-B2b time comparison from DoY 163 to 171 in 2022. The results of the time comparison reflect the TAI retention of the NTSC and TL laboratories. Certainly, the time comparison link error mainly contains the local receiver clock error and PPP solution error with precise satellite products and models. Taking the GBM PPP time comparison as a reference, the time transfer result difference is shown in Fig. 6 and Fig. 7. Meanwhile, Table 4 lists the STD values of the residuals. The fluctuation of the curve reflects the time comparison error of different PPP-B2b models. Whether NTSC-TLM2 or NTSC-USUD time link, QF has some improvement in accuracy and the residuals of time comparison really fluctuate within 1 ns. Compared with the DF PPP-B2b time transfer, the STD value of the long-baseline time comparison using QF PPP-B2b time transfer model is improved from 0.25 ns to 0.2 ns. The truth that the QF PPP-B2b model makes the time error of the stations involved in the time comparison within 1 ns can be demonstrated.

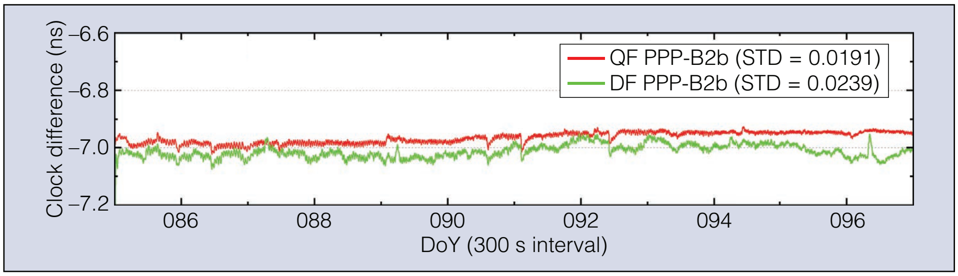

Fig. 4. QF and DF PPP-B2b zero-baseline CCD.

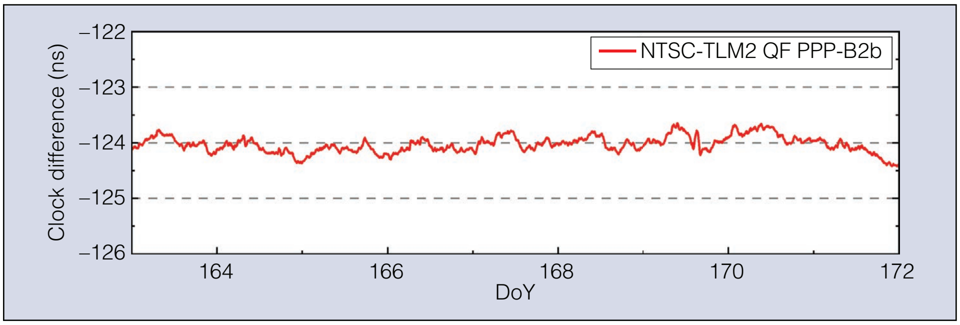

Fig. 5. QF PPP-B2b long-baseline time comparison for 9 d from DoY 163 to 171 in 2022.

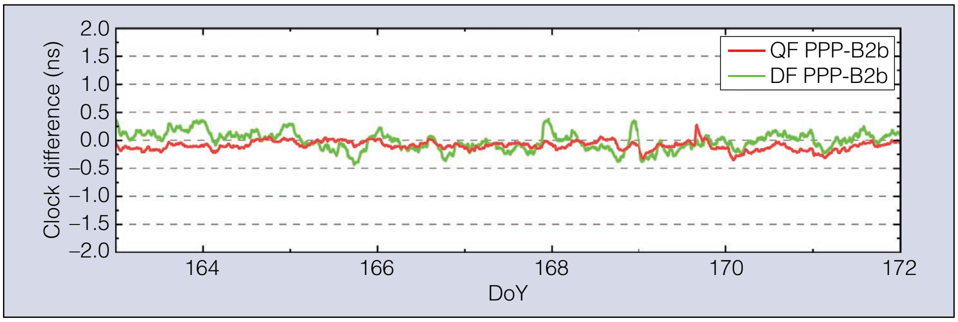

Fig. 6. Difference time sequences of NTSC-TLM2 PPP time comparison results using the GBM product and the PPP-B2b product.

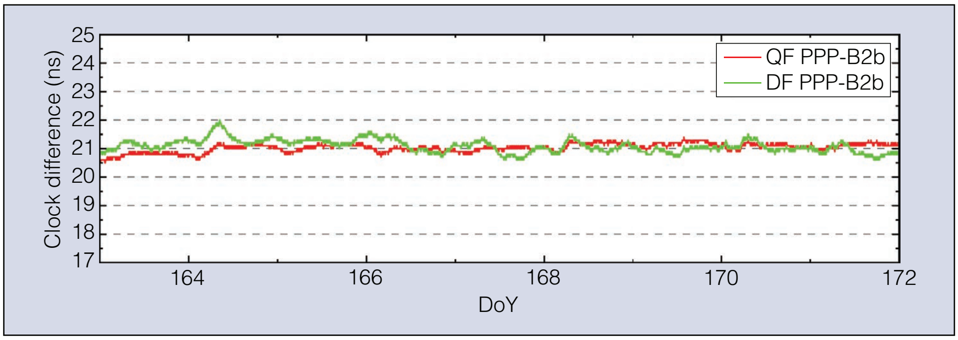

Fig. 7. Difference time sequences of NTSC-USUD PPP time comparison results using the GBM product and the PPP-B2b product.

Table 4 – STD values of PPP-B2b time comparison residual

Time link	Model	
	QF PPP-B2b	DF PPP-B2b
NTSC-TLM2	0.161	0.215
NTSC-USUD	0.201	0.255

Conclusions

PPP time transfer accuracy can reach sub-nanosecond without limitations on distance. Conversely, PPP execution requires the external provision of real-time precise satellite products. Since August 2020, BDS-3 has provided precise products called the PPP-B2b product through three GEO satellites instead of a network, which solves the problem of interruption caused by network fluctuation in the field of real-time PPP time comparison. The truth that BDS-3 DF PPP-B2b time transfer can reach an accuracy of sub-nanosecond has been proven. Considering the BDS-3 can provide users with a wide range of frequency signals for PPP and the increasing observation information will accelerate the PPP convergence and improve the accuracy of time transfer, the QF PPP-B2b time transfer was studied.

A QF PPP-B2b time transfer model was proposed and validated from zero-baseline CCD and long-baseline time links. To ensure the accuracy of the results, two TAI keeping laboratories (NTSC and TL) and the IGS external atomic clock station (USUD) in the Asia-Pacific region were selected. To make full use of the BDS-3 frequency signals to verify the accuracy of QF PPP-B2b time transfer model, the newly broadcasted B1C and B2a signals of BDS-3 were used alongside existing B1I and B3I signals. The results showed that QF PPP-B2b time transfer had smoother CCD results and fluctuated within 0.1 ns, compared to the DF PPP-B2b model. In the NTSC-TLM2 long-baseline time comparison experiment, the QF PPP-B2b time comparison residuals fluctuate within 1 ns. Meanwhile, taking the GBM PPP time transfer as a reference, whether NTSC-TLM2 or NTSC-USUD long-baseline time link, QF has some improvement in accuracy and the residuals of time comparison really fluctuate within 1 ns. According to the STD values calculated, QF has a significant improvement compared to DF PPP-B2b time transfer. In conclusion, the truth that the QF PPP-B2b model makes the time error of the stations involved in time comparison within 1 ns can be demonstrated.

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